# SAMPLE

## Cosmic Explorers: Space Navigation

Finding Your Way Across the Solar System

Aira Astralis

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# Introduction

Navigation is a field of study that focuses on the process of monitoring and controlling the movement of a craft or vehicle from one place to another. The field of navigation includes four general categories: land navigation, marine navigation, aeronautic navigation, and space navigation. It is also the term of art used for the specialized knowledge used by navigators to perform navigation tasks. All navigational techniques involve locating the navigator's position compared to known locations or patterns.

Navigation, in a broader sense, can refer to any skill or study that involves the determination of position and direction. In this sense, navigation includes orienteering and pedestrian navigation. For marine navigation, this involves the safe movement of ships, boats and other nautical craft either on or underneath the water using positions from navigation equipment with appropriate nautical charts (electronic and paper). Navigation equipment for ships is mandated under the requirements of the SOLAS Convention, depending on ship size.

For land navigation, this usually involves the movement of persons, animals and vehicles from one place to another by means of navigation equipment (such as a compass or GNSS receivers), maps and visual navigation marks across urban or rural environments. Aeronautic (air) navigation involves piloting an aircraft from one geographic position to another position while monitoring the position as the flight progresses. The term stems from the 1530s, from Latin navigationem (nom.

Polynesian navigation is probably the earliest form of open-ocean navigation; it was based on memory and observation recorded on scientific instruments like the Marshall Islands Stick Charts of Ocean Swells. Early Pacific Polynesians used the motion of stars, weather, the position of certain wildlife species, or the size of waves to find the path from one island to another.

# Chapter 01: The Discovery of Space Navigation

Although land astrolabes were invented in the Hellenistic period and existed in classical antiquity and the Islamic Golden Age, the oldest record of a sea astrolabe is that of Spanish astronomer Ramon Llull dating from 1295. The perfecting of this navigation instrument is attributed to Portuguese navigators during early Portuguese discoveries in the Age of Discovery.

The earliest known description of how to make and use a sea astrolabe comes from Spanish cosmographer Martin Cortes de Albacar's Arte de Navegar (The Art of Navigation) published in 1551, based on the principle of the archipendulum used in constructing the Egyptian pyramids. However, the first altitude measuring instrument to navigate extensively used at sea was the quadrant. This was reintroduced by Leonardo of Pisa in the 13th century. Its first recorded use was in 1461 by Diogo Gomes.

As well as astrolabes and quadrants, the first cross-staff used in navigation was known from the 14th century onwards, believed to have come from early Arab navigators. However, it had many errors and was also difficult to use as it required squinting at the sun. These disadvantages were overcome with the invention of the backstaff in 1595 by John Davis. Widespread open-seas navigation using the astrolabe, quadrant, backstaff and the compass started during the Age of Discovery in the 15th century.

The Portuguese began systematically exploring the Atlantic coast of Africa from 1418, under the sponsorship of Prince Henry. In 1488 Bartolomeu Dias reached the Indian Ocean by this route. In 1492 the Spanish monarchs funded Christopher Columbus's expedition to sail west to reach the Indies by crossing the Atlantic, which resulted in the Discovery of the Americas. In 1498, a Portuguese expedition commanded by Vasco da Gama reached India by sailing around Africa, opening up direct trade with Asia.

Soon, the Portuguese sailed further eastward, to the Spice Islands in 1512, landing in China one year later. The first circumnavigation of the earth was completed in 1522 with the Magellan-Elcano expedition, a Spanish voyage of discovery led by Portuguese explorer Ferdinand Magellan and completed by Spanish navigator Juan Sebastian Elcano after the former's death in the Philippines in 1521. For sailing ships, other developments took place with charting and methods to record courses.

One of the oldest surviving marine charts is the Carta Pisana, drawn on a sheepskin, dating to 1275. On land, improvements in the production of maps led to improved navigation by armies, traders and other travellers. For sailing ships, navigation by dead reckoning requires frequent recording of course

changes and the ship tacks with the wind.

To prevent paper charts, which were expensive and in the early days, rare, from being worn out, other methods were used, including the Traverse board and traverse tables (the oldest traverse tables, dates back to 1428). Quadrants were further developed by inventors such as Robert Hooke, Isaac Newton and John Hadley leading to the invention of the octant. Developments in mathematics were also important in the history of navigation.

These include initially meridional parts, then developments in spherical trigonometry and logarithms enabled navigators from the 1700s onwards to navigate more accurately. On land, mathematical and new instruments led to developments in Surveying and triangulation which further improved maps, as well as the construction of better roads, paths, canals and eventually railways. Development of an accurate marine chronometer under John Harrison and others ensured accurate timekeeping for calculating longitude.

Further improvements in ocean navigation led to the first proper sextant in 1757, the parts and usage developed by various inventors including Pierre Vernier and John Campbell. Various methods for calculation with sextant and chronometer evolved over time, beginning with the Duller method (1728) but reached their most accessible with the Douwes method (1821), the Sumner method (1837), modified by Henry Raper (1844) and the Marc St Hilaire or intercept method (1877).

Modifications to the magnetic compass and better methods of determining course were also important, include developments in the compass by Matthew Flinders, Lord Kelvin and others. The sextant, together with the chronometer, compass and astronomical calculations became the most widely used methods of maritime navigation until developments in the 20th century with radio-navigation and gyrocompasses. These in turn were superseded with the advent of computers, electronic calculators and later satellite navigation in the 20th century.

On land, the development of handheld GPS occurred in the 1980s and with the advent of smartphones, with in-build compassess and satellite receivers, navigation is now widely achieved through technology globally. In terrestrial navigation, the location of a person, ship, plane, etc is defined as a position using a reference point/coordinates (see Cartesian coordinate system). Positions can either be referenced as latitude/longitude or a distance and direction from a fixed reference point (bearing).

Lines of position can be derived from a variety of methods and equipment. By determining and monitoring positions it is possible to find and direct a person,

ship, plane, etc in a scientific way from one place to another. This often involves the use of maps or charts from which if desired, courses can be calculated or followed depending on the projection or methods used (Rhumb line, Great circle, etc).

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# ABOUT THE AUTHOR

Aira Astralis is a science writer specializing in astronomy and space exploration. Her work spans the full breadth of humanity's relationship with the cosmos -- from the earliest naked-eye observations to the engineering marvels of modern space missions.

Her books are built on a foundation of primary sources: peer-reviewed research, mission archives, and direct engagement with the scientific community. She writes for readers who want to understand not just what we know about the universe, but how we came to know it.

She believes that the most important frontier is not out there but in the gap between what scientists discover and what the public understands. Her books exist to close that gap.